ANDROID APP DEVELOPMENT · TASK 2

Dynamic Employee Directory

Advanced Kotlin, OOP, RecyclerView, Fragments, View Binding, Navigation Component and Material Design 3

9	17	15	M3
screens	tests passing	Kotlin files	Material Design

DEVELOPER
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PROGRAMME
Sumerix Global · Android App Development · 36 weeks

BATCH START
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STACK
Kotlin 2.0 · AGP 8.7 · Material 3 · minSdk 26 / target 35

1 · Overview

Advanced Kotlin, OOP, RecyclerView, Fragments, View Binding, Navigation Component and Material Design 3. Package `com.eswar.employeedirectory`.

- Single Activity + 6 Fragments (Dashboard, Employees, Detail, Departments, Company, Contact) on one navigation graph with Safe Args (`departmentName`, `employeeId`)
- Bottom navigation + Top App Bar; back stack handled by `NavController`
- RecyclerView with `ListAdapter` + `DiffUtil` and multiple view types (department header + employee card), grid on tablets
- Live search box and department filter chips; employee detail with skills chips, call / e-mail actions and a contact dialog
- OOP model: `Searchable` interface, abstract `Person` (abstract `roleSummary()`), `Employee` data class (polymorphic `matches()`), `Department` enum, sealed `DirectoryItem`
- Advanced Kotlin: higher-order functions, lambdas, collections (groupBy, associate, sumOf), extension and scope functions
- View Binding everywhere, with `_binding = null` in `onDestroyView()`; dark theme; Snackbar, FAB and dialog components

9
screens captured

17
automated tests · 0 failing

15
Kotlin source files

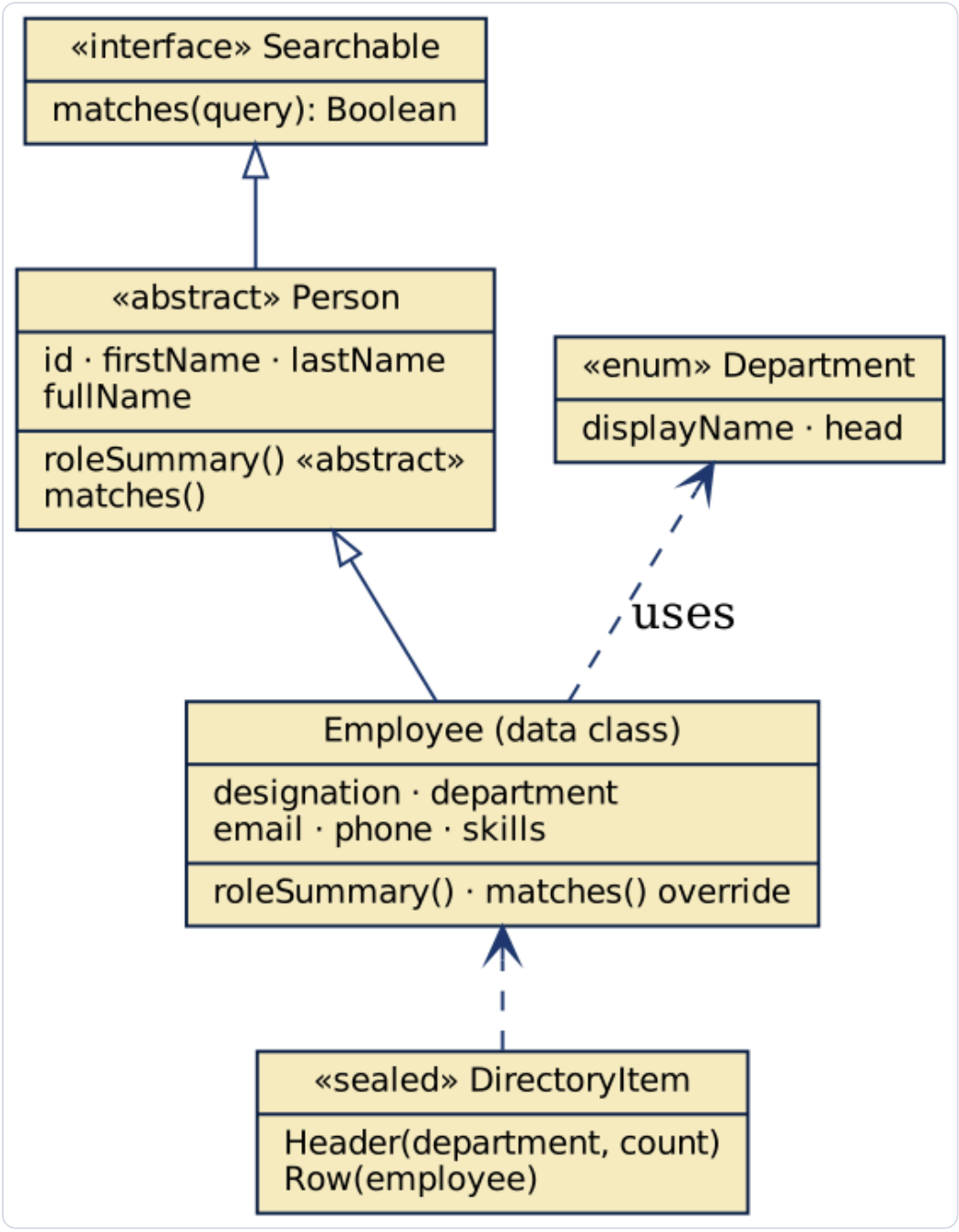
17
XML layouts

2 · Brief coverage

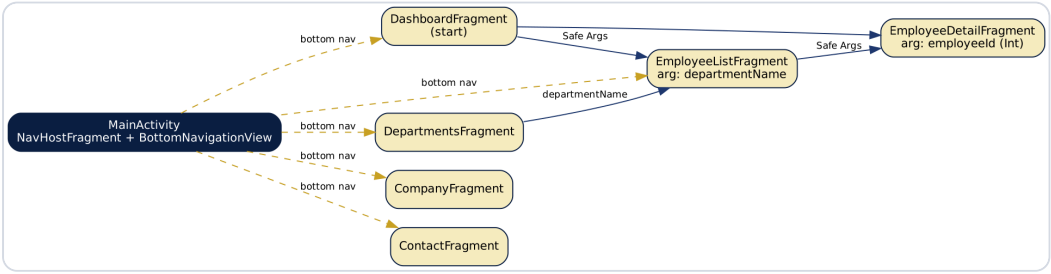
Brief part	Implementation
A – Advanced Kotlin	model/EmployeeExtensions.kt, data/EmployeeRepository.kt (collections, lambdas, scope functions)
B – OOP	model/Person.kt (interface + abstract class), Employee.kt (inheritance, polymorphism), Department.kt (enum)
C – Views	Cards, RecyclerView, NestedScrollView, Toolbar, FAB, Chip, Bottom Navigation
D – RecyclerView	ui/adapter/DirectoryAdapter.kt (2 view types, DiffUtil, click lambdas), DepartmentAdapter.kt
E – Fragments	6 Fragments; data passed with Safe Args; lifecycle-safe binding
F – View Binding	viewBinding = true; no findViewById in the codebase
G – Navigation Component	res/navigation/nav_graph.xml, NavController + BottomNavigationView, back stack
H – Material Design 3	Top App Bar, Bottom Navigation, cards, chips, FAB, Snackbar, dialog
I – Dynamic UI	24 fictional employees, 6 departments, responsive grid (list_span 1 phone / 2 tablet)

J/K – Mini project & challenge	Dashboard, list, details, departments, company info, contact, bottom nav

3 · Architecture



Class diagram

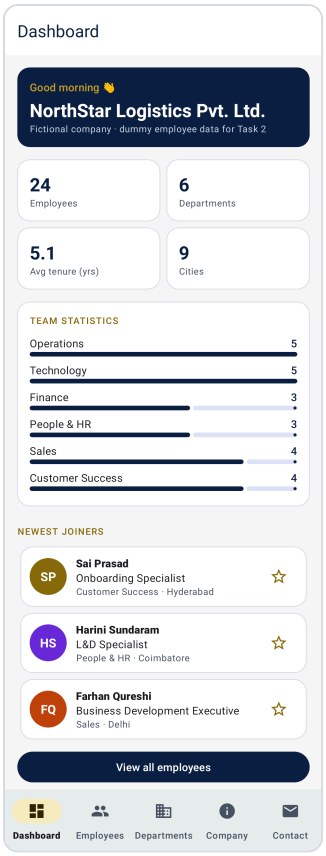


Navigation graph

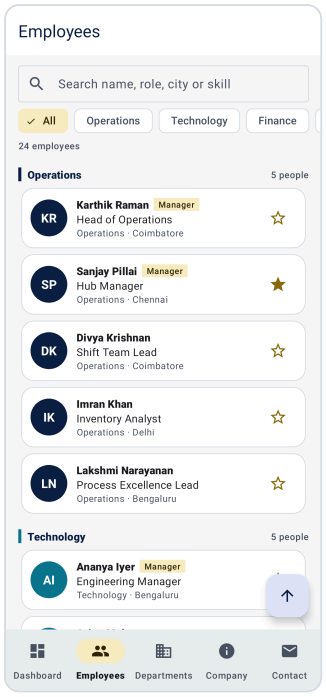


Recyclerview flow

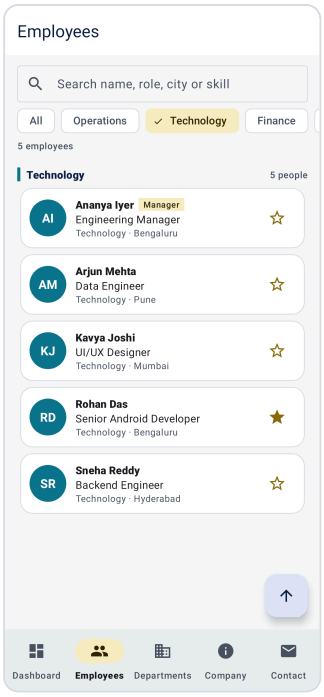
4 · Screens (phone, light & dark)



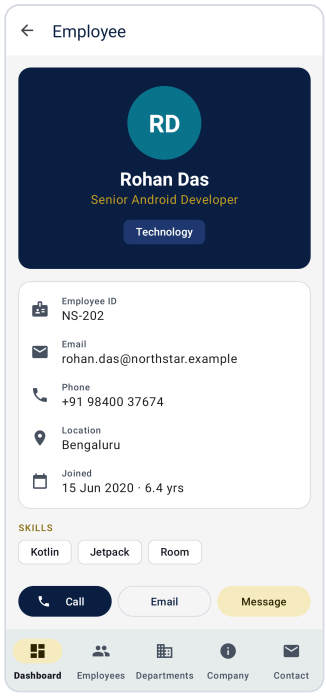
Dashboard



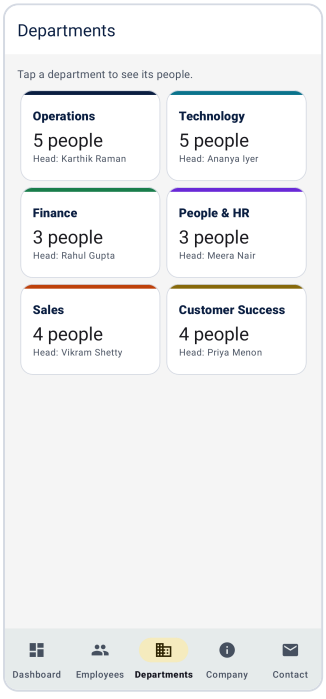
Employee list



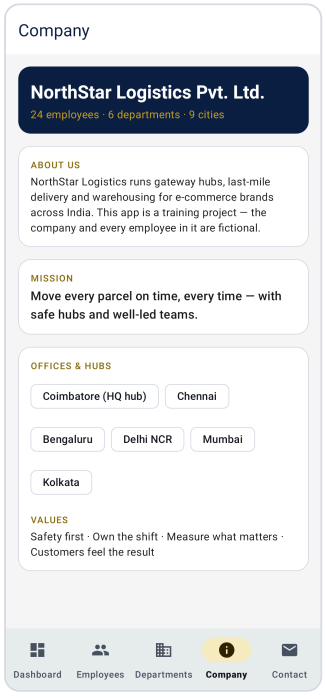
Filtered technology



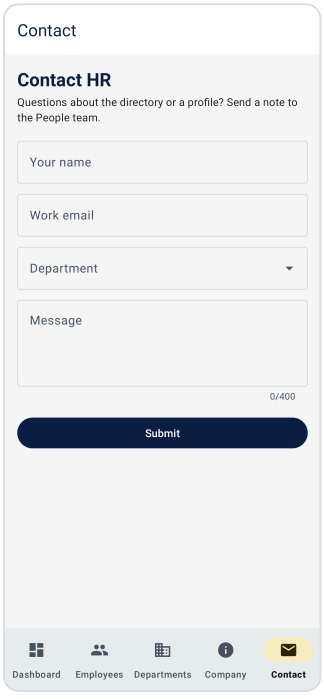
Employee detail



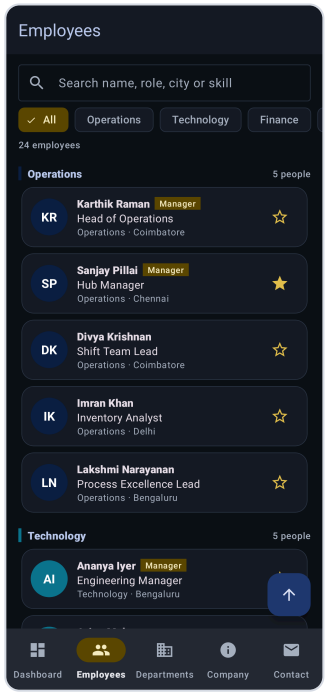
Departments



Company

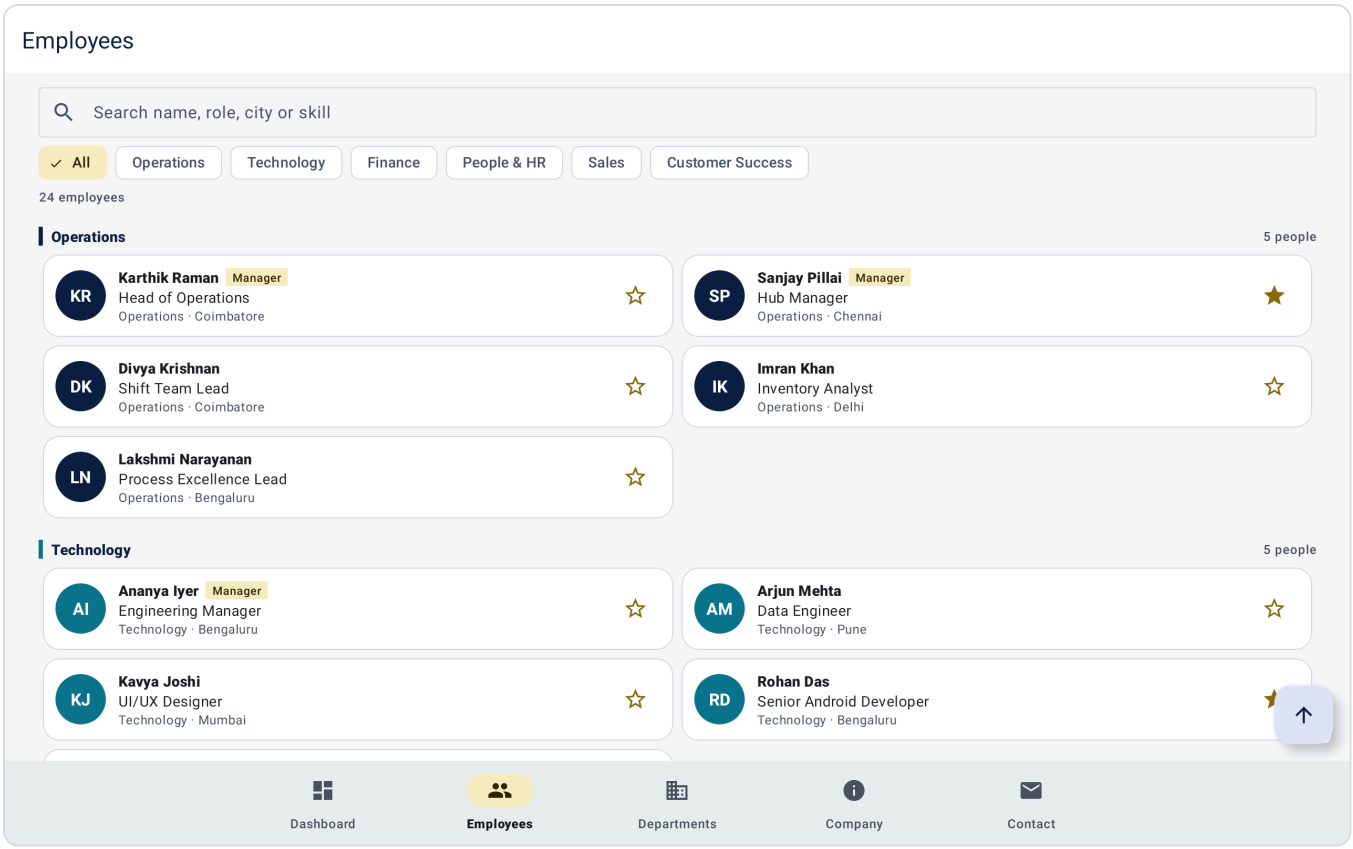


Contact



Employee list dark

5 · Tablet layout (sw600dp)



Tablet two column list

6 · Testing

Test class	Tests	Failures	Cases
EmployeeLogicTest	5	0	searchMatchesNameRoleCityAndSkill, favouriteToggleIsReversible, repositoryHasTwentyFourEmployeesInSixDepartments, extensionsAndTenure, departmentFilterAndSectionHeaders
NavigationTest	3	0	tappingAnEmployeeOpensDetailWithSafeArgs, startsOnDashboardAndBottomNavSwitchesFragments, departmentArgumentFiltersTheList
ScreenshotTest	9	0	s06_company, s03_filtered, s02_employees, s04_detail, s08_dark, s05_departments, s09_tablet, s07_contact, s01_dashboard

Tests run on the JVM with Robolectric (API 34). Screenshot tests render every screen with Roborazzi in native graphics mode, so the images in this document are produced by the real XML layouts, themes and resources.

7 · Project structure

```
src/
  main/
    AndroidManifest.xml
    java/
      com/
        eswar/
          employeedirectory/
            data/
              EmployeeRepository.kt
            model/
              Department.kt
              Employee.kt
              EmployeeExtensions.kt
              Person.kt
            ui/
              MainActivity.kt
              UiExtensions.kt
              adapter/
                DepartmentAdapter.kt
                DirectoryAdapter.kt
              company/
                CompanyFragment.kt
              contact/
                ContactFragment.kt
              dashboard/
                DashboardFragment.kt
              departments/
                DepartmentsFragment.kt
              detail/
                EmployeeDetailFragment.kt
              list/
                EmployeeListFragment.kt
    res/
      drawable/
        bg_avatar_circle.xml
        ic_add.xml
        ic_apartment.xml
        ic_arrow_up.xml
        ic_badge.xml
        ic_calendar.xml
        ic_chevron.xml
        ic_cloud_off.xml
        ic_dashboard.xml
        ic_delete.xml
        ic_edit.xml
        ic_info.xml
        ic_launcher_foreground.xml
        ic_location.xml
        ic_mail.xml
        ic_payments.xml
        ic_people.xml
        ic_phone.xml
        ic_search.xml
        ic_sort.xml
        ic_star.xml
        ic_star_border.xml
      layout/
        activity_main.xml
        fragment_company.xml
        fragment_contact.xml
        fragment_dashboard.xml
        fragment_departments.xml
```

```
        fragment_employee_detail.xml
        fragment_employee_list.xml
        item_department.xml
        item_department_header.xml
        item_employee.xml
        item_info_row.xml
        item_team_stat.xml
        view_filter_chip.xml
        view_skill_chip.xml
        view_stat_card.xml
        view_text_card.xml
        view_text_card_mission.xml
      menu/
        bottom_nav.xml
      mipmap-anydpi-v26/
        ic_launcher.xml
      navigation/
        nav_graph.xml
      values/
        colors.xml
        dimens.xml
        ic_launcher_background.xml
        strings.xml
        themes.xml
      values-night/
        colors.xml
      values-sw600dp/
        dimens.xml
  test/
    java/
      com/
        eswar/
          employeedirectory/
            EmployeeLogicTest.kt
            ui/
              NavigationTest.kt
              ScreenshotTest.kt
    resources/
```